GEK-36487

INSTRUCTIONS

RADIANT-HEAT DETECTOR SYSTEM

3S7505HR500 CONTROL

3S7505HS500 SCANNER

GENERAL ELECTRIC

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These instructions do not purport to cover all details or variations in equipment nor to provide for every possible contingency to be met in connection with installation, operation or maintenance. Should further information be desired or should particular problems arise which are not covered sufficiently for the purchaser's purposes, the matter should be referred to General Electric Company.

INTRODUCTION

This Radiant-Heat Detector System consists of a scanner which supplies a chopped signal to a transistorized amplifier control. The amplifier control has a choice of optional output devices which consist of a standard plug-in DPDT relay, an AC static switch (plug-in module), a mercury-wetted relay, or a built-in logic voltage output.

This system is primarily used in, but not limited to, steel mill applications for sensing the presence or absence of the hot material throughout its process of manufacture. Some of the various applications may include use in blooming and slabbing mills, coilers, finishing mills, rod and bar mills, and in furnace areas.

The temperature range of the material which can be scanned will vary depending upon the background temperature, the emissivity of the hot material, and the percent of viewed area filled.

DESCRIPTION

This system consists of the following components:

Control Panel	3S7505HR500
Scanner Body	3S7505HS500
Alternate Scanning Assembly (Body must be equipped with one or the other)	3S7505KL502 Lens Assembly with Extension Barrel
	3S7505KL503 Wide Angle Assembly
Optional Output Devices (Logic output, if required, is built in the control panel and obtained by making the required jumper connections on the terminal board.)	3S7505KH501 DPDT Relay
	3S7505KS503 Static Switch
	3S7505KW501 Mercury Wetted Relay
Scanner Connection Cable: for temperature below 175°F	44A310526-001 2-Conductor, Shielded
for temperature above 175°F up to a maximum of 400°F	44A317706-001 2-Conductor, Shielded

The 3S7505HR500 Control Panel (Figure 1) is a transistorized amplifier with adjustable sensitivity and time delay controls. It contains a light to indicate the presence or absence of hot material. Terminal boards provide connections

DESCRIPTION (continued)

for power input, ground, output device (relay, logic or static switch), optional time delay, and scanner.

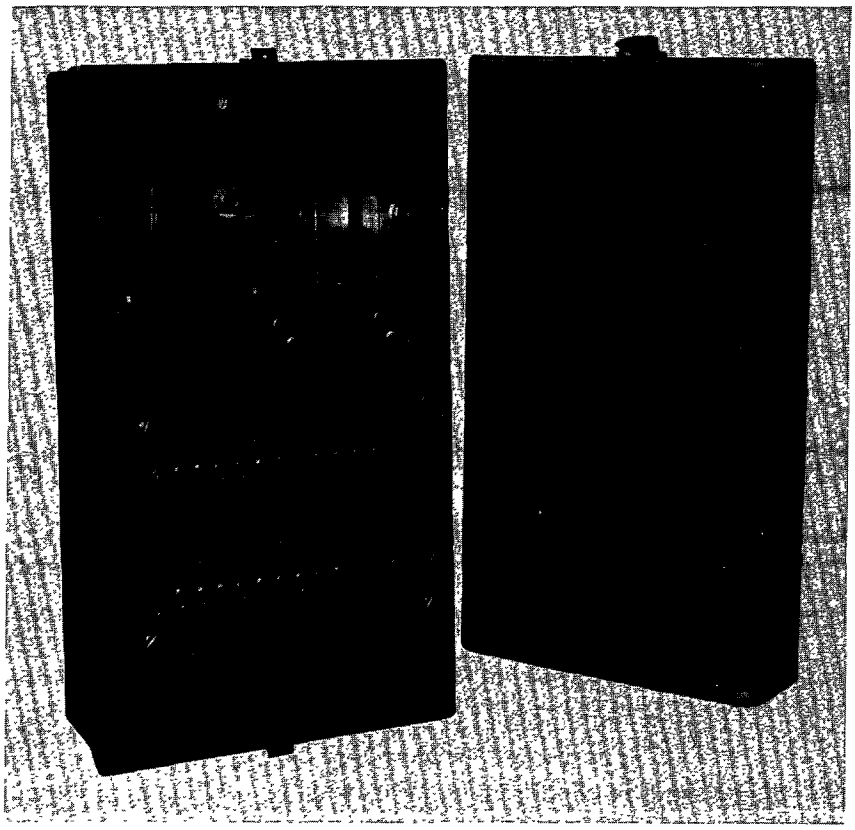


Figure 1. Radiant-Heat Detector Control 3S7505HR500

The 3S7505HS500 Scanner Body is a rugged unit consisting of a motor-driven chopper, heat sensing lead-sulfide detector, coolant chamber with fittings, and adjustable mounting base. The base allows aiming adjustment of $\pm$ 30° horizontally and $\pm$ 20° vertically.

The body must be equipped with one of the alternative scanning assemblies described as follows:

(1) A 3S7505KL502 Lens with extension barrel (Figure 2) which provides for scanning a restricted field of view. It is supplied with two apertures in

DESCRIPTION (continued)

addition to the regular lens opening, increasing its versatility. The extension barrel, which provides environmental protection for the lens assembly, can be removed and replaced with an air purge adaptor*, catalog number 3S7505KF139.

(2) A 3S7505KL503 Wide Angle Scanning Assembly (Figure 3) which is supplied with two sizes of windows extending the scope of wide area scanning. It is equipped with an air purge adaptor used to aid in the prevention of dust, dirt, fog, steam, etc., from accumulating on the viewing window.

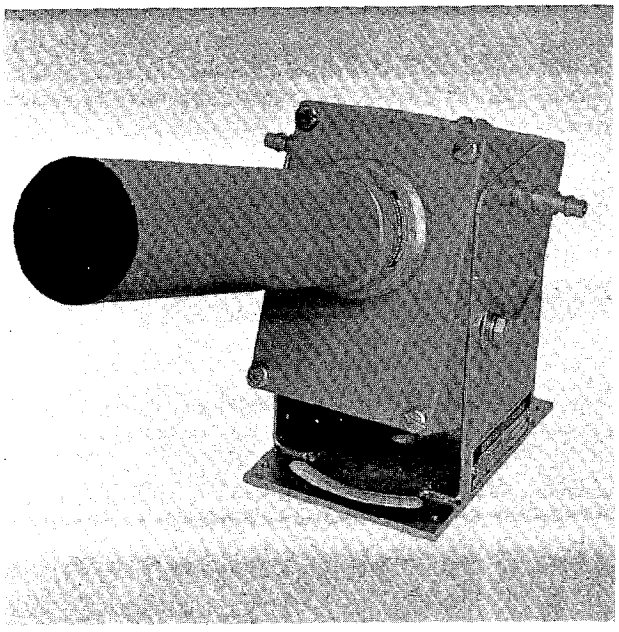


Figure 2. Scanner, 3S7505HS500 Equipped With 3S7505KL502A1 Lens Assembly

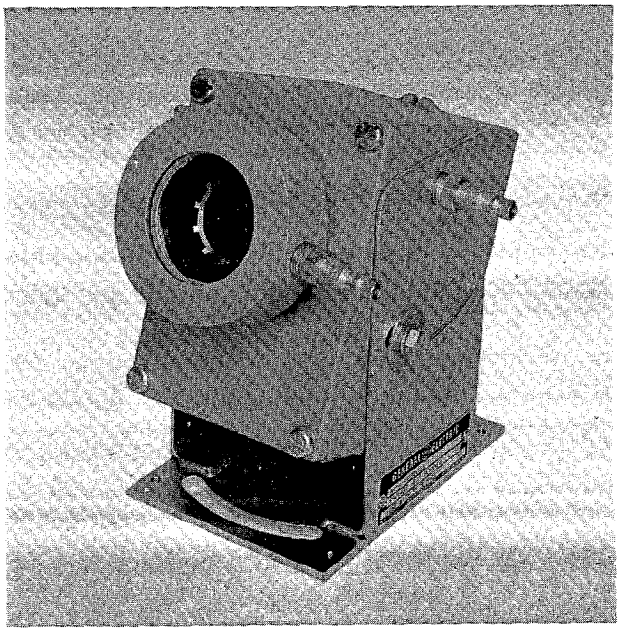


Figure 3. Scanner, 3S7505HS500 Equipped With 3S7505KL503A1 Wide Angle Scanning Assembly

* The extension barrel must be removed if the air purge adapter is used on the 3S7505KL502.

RECEIVING, HANDLING AND STORAGE

Immediately upon receipt, carefully unpack the equipment and examine it for any damage sustained in transit. If injury or rough handling is evident, a damage claim should be filed immediately with the carrier and the nearest General Electric sales office should be notified.

The control panel and scanner equipment should be stored in a clean, dry place until ready for installation.

SPECIFICATIONS

Input Power-Control 3S7505HR500
Models B1, B6 - 115/230 VAC ± 10% 50/60 Hz, 7VA

Input Power-Scanner 3S7505HS500

Model C6 - 115 VAC ± 10% 50/60 Hz, 7VA

Optional Output Devices (See Table 1)

These devices are supplied separately to be plugged in by the user, except the logic output which is built in the control panel.

Logic Output

24 volt DC logic level with 680 ohm pull up resistor. Response time 0.003 seconds.

Ambient Temperature

Control 0°F to 135°F
Scanner 0°F to 135°F*

Enclosures

NEMA 3/12 weather resistant/dust tight/oil and coolant resistant.

Time Delay (Built-In)

Adjustable 0.1 to 1.0 second with time delay on pickup of relay upon removal of heat signal.

Mode of Operation (See Figure 8)

Relay deenergizes on presence of heat signal.

*For ambient temperatures exceeding 135° F up to 400° F maximum at the scanner, see discussion entitled COOLING REQUIREMENTS.

SPECIFICATIONS (continued)

Mode of Operation (continued)

Connections are provided to seal in the relay on pickup - externally reset.

Field of View

Orientation	long dimension is vertical (perpendicular to axis of cooling fittings)
3S7505KL502 . . . . . Lens Assembly (with or without aperture)	0.2 in. X 1.0 in. per foot of scanning distance
3S7505KL503 Wide Angle Scanning Assembly:	
(1) 0.01 in. X 1.0 in. window . . .	0.2 in. X 10 in. per foot
(2) 0.06 in. X 0.17 in. window . .	0.7 in. X 2.0 in. per foot
Operating Characteristics	Object temperature vs background temperature and selection of scanning assembly with proper apertures and windows (see curves A, B and C).

TABLE I

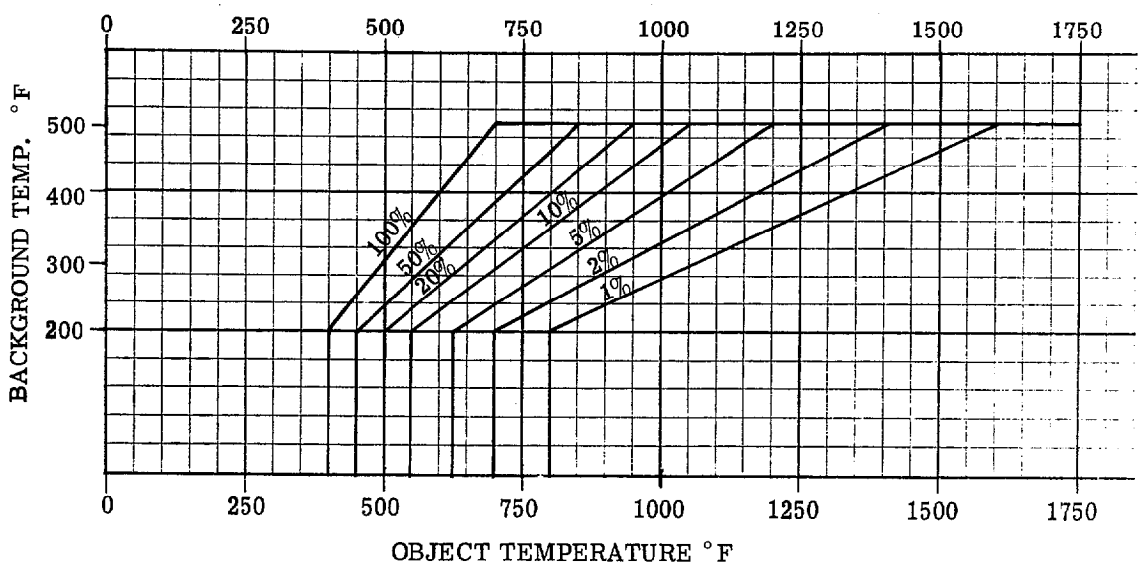
CHARACTERISTICS OF THE 3S7505HR500

WHEN USED WITH OPTIONAL OUTPUT DEVICES

When Used with:	Hermetic Relay 3S7505KH501A1	Mercury-Wetted Contact Relay* 3S7505KW501A1	Static Switch 3S7505KS503A1
Total Response Time To Energize	0.018 second	0.009 second	See specification supplied for Static Switch per GEK-14876
To Deenergize	0.018 second	0.009 second	Note: Allow an additional 0.003 sec response time to the time specified in GEK-14876.
Output	10A, DPDT, Resistive, 230 V AC or 28 V DC	*125 VA, AC, 3.5 Amps, max; *250 VA, DC, 5 Amps Max; 300 V Max.	
Form	C (Break before make)	D (bridging: make before break)	
Operations at rated load	100,000	1 billion	
Mechanical Life	500 million	1 billion	

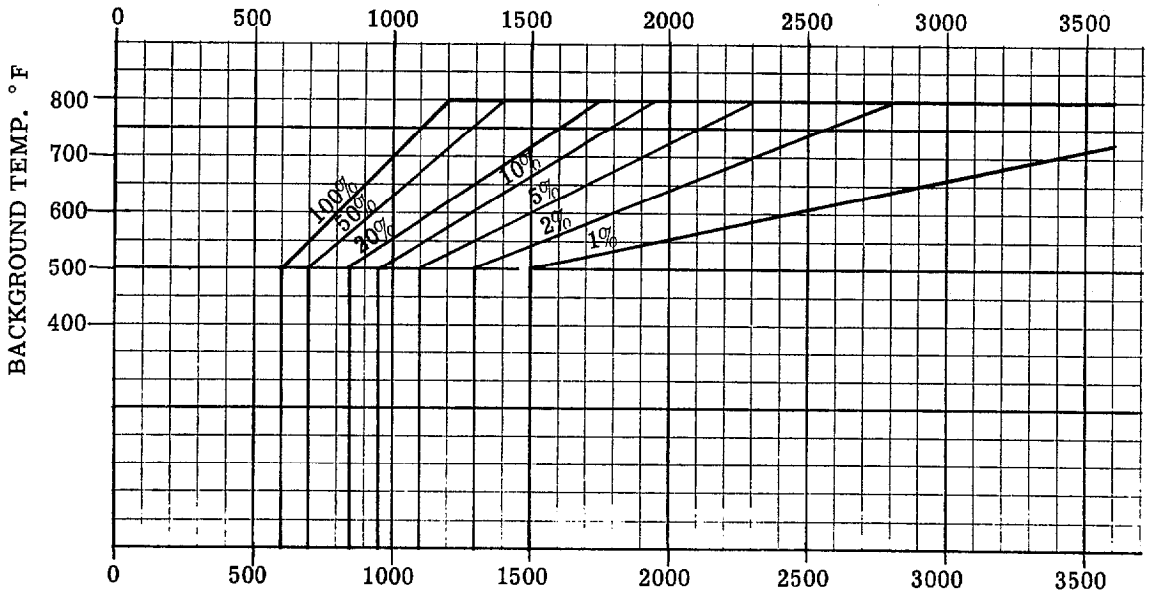
*Resistance - capacitance protection must be across each contact used (see GEI-98299). Upright mounting of relay must be used. For relay with attached plug, cable and socket (in right-angle bracket) for separate mounting, order 3S7505KW501B1.

*See Table 3



Family of Curves Representing (Emissivity* of Material)X(Percent Field of View Filled) for the scanner Body Equipped with the 3S7505KL502 Lens Assembly with no Aperture.

CURVE "A"

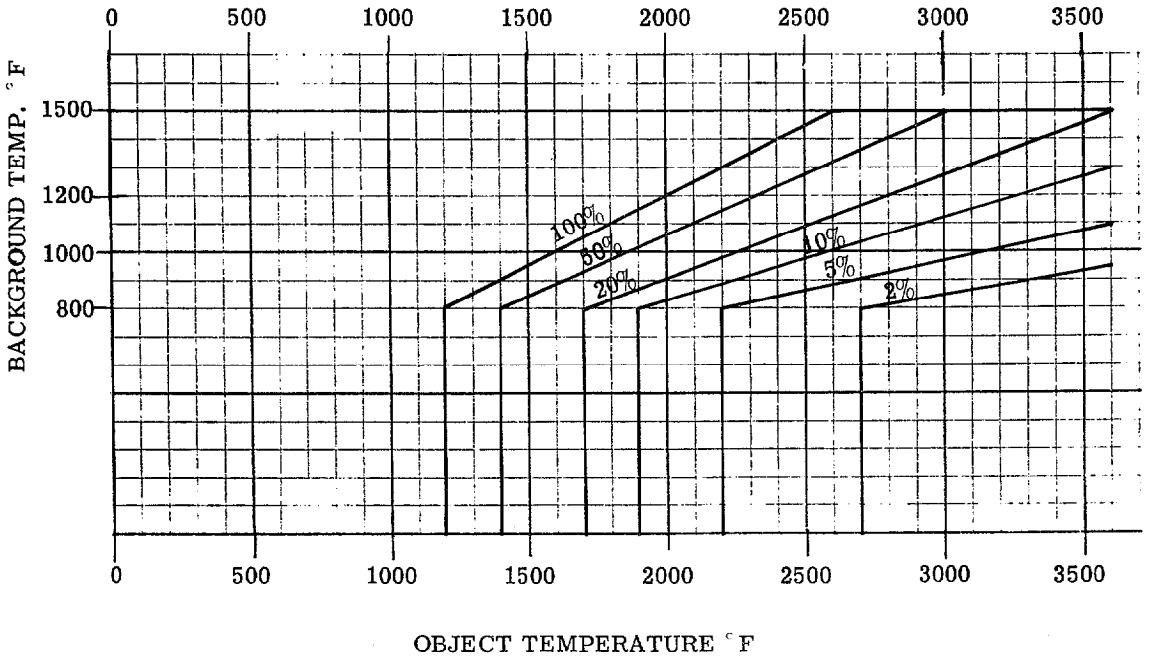


Family of Curves Representing (Emissivity* of Material)X(Percent Field of View Filled) for the Scanner Body Equipped with the Following Alternatives:

(1) 3S7505KL502 Lens Assembly with 0.175 in. diameter aperture.
(2) 3S7505KL503 Wide Angle Assembly with 0.01 in. X 1 in. window.
(3) 3S7505KL503 Wide Angle Assembly with 0.06 in. X 0.167 in. window.

*See Table 3

CURVE "B"



Family of Curves Representing (Emissivity* of Material)X(Percent Field of View Filled) for the Scanner Body Equipped with 3S7505KL502 Lens Assembly with 0.031 in. Diameter Aperture.

*See Table 3

CURVE "C"

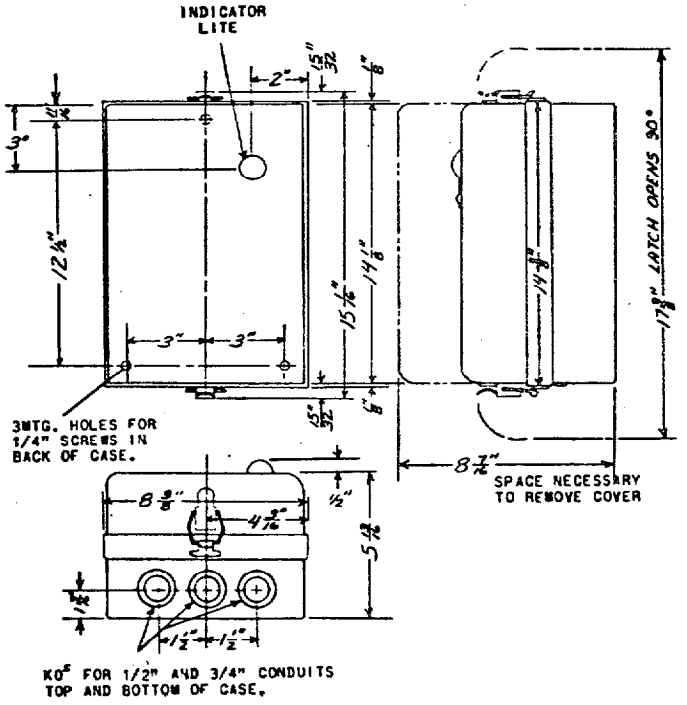


Figure 4. Outline Diagram of 3S7505HR500 Control (44A300232 Rev. 0)

INSTALLATION

MOUNTING

Scanner

A thorough investigation of the application is required before mounting the scanner. A careful selection of the scanner location can result in less stringent cooling and heat shielding requirements, as well as easier and less frequent maintenance and a longer trouble-free scanner life.

CAUTION

NOTE THAT THE SCANNER MUST NOT BE EXPOSED TO THE MILL ENVIRONMENT OR HIGH TEMPERATURES UNTIL THE LENS ASSEMBLY HAS BEEN INSTALLED AND PROPER COOLING AND POWER HAS BEEN APPLIED. PREMATURE EXPOSURE MAY CAUSE DAMAGE, EVEN THOUGH NO POWER IS APPLIED TO THE SCANNER.

The distance from the material to be scanned to the scanner is dictated by the application (refer to APPLICATION section); however, in general, large hot material will permit greater distances. This is fortunate because these targets usually present scanner heating problems. To avoid heating when scanning this type object, the scanner must not be mounted directly above or in close proximity to the material. Instead, it should be located off to one side and as far away as is practical -- within the limits imposed by the application. When dealing with large hot materials, it is also important to use the smallest lens aperture which is commensurate with the application.

Never locate the scanner where it will be subjected to high temperatures unnecessarily. Avoid niches which may trap heat. Install reflective shields to deflect radiant heat away from the scanner rather than shields or shrouds which will absorb radiant energy. If necessary, install thermal insulation between the scanner and a hot mounting surface.

Always select a scanner location in which the coolest possible background surface is in the scanner field of view when the hot object is not present. Refer to the performance curve (A, B or C) being used for maximum permissible background temperature. Also, avoid direct or reflected sunlight or incandescent illumination in the scanner field of view. Finally, the scanner should always be aimed with a minimum downward angle of 5° from the horizontal to prevent dirt and debris from falling or collecting on the lens. The air purge adaptor is also helpful in keeping the lens clean.

Scanner Assemblies

The application for which this equipment is intended determines which scanning assembly is to be used with the scanner body and if an aperture or wide angle

INSTALLATION (continued)

MOUNTING (continued)

Scanner Assemblies (continued)

window slot is required. If this has not been determined, refer to the section entitled APPLICATION for detailed discussion.

The following steps describe the method of attaching the scanning assembly to the scanner body, and installation of the aperture in the lens assembly or the window slot in the wide angle unit.

A. 3S7505KL502 Lens Assembly

(1) Insert the selected aperture, if required for the application, on the lens retaining ring (end opposite the extension barrel) and retain it with the additional ring supplied with the apertures.

(2) Remove the 2-inch diameter protective plug from the scanner body face.

(3) Screw the lens barrel assembly to the scanner body until fully seated.

(4) If the air purge adaptor, catalog number 3S7505KF139, is used in place of the extension barrel, unscrew the extension barrel from the lens assembly and screw the adaptor on until snug; then back off 1/2 turn. Tighten set screw provided in collar. Connect filtered shop air to the 1/4-inch fitting. (See additional information in B (7) below for this air purge adaptor.

B. 3S7505KL503 Wide Angle Scanning Assembly

One of the two window slots supplied with this assembly must be used. Also, the window slot must be oriented in the scanner assembly to align the long axis of the slot with the long dimension of the detector cell and chopper disk slots, which are vertical.

(1) Remove the 2-inch diameter protective plug from the scanner body face.

(2) Screw the wide angle scanning assembly into the body until seated and mark the assembly in order to identify the vertical orientation.

(3) Remove the scanning assembly from the scanner body.

(4) Insert the selected window slot (with the long dimension vertical as determined in step (2) above) in the end opposite the glass window and press in the retaining ring supplied with the windows.

(5) Again screw the scanning assembly into the body until fully seated.

INSTALLATION (continued)

MOUNTING (continued)

Scanner Assemblies (continued)

(6) Check that the slot is vertical and make adjustment if necessary.

(7) The adaptor for air purging to keep the window clean contains a 1/4-inch quick disconnect plug. It may be rotated one-quarter turn in either direction for accessibility to the air supply by loosening the locking set screw. Filtered shop air from 50 to 100 psi is recommended for this air purge adaptor. The use of a Perfecting Service Company Catalog Number F100A or equivalent quality filter is recommended. The load on the air supply will be approximately 1.5 cu. ft./min. at 50 psi.

Mount the scanner (see outline-Figure 5) at the desired location which provides the required field of view (see SPECIFICATIONS). The scanning assembly used, i.e., either lens or wide angle with window slot, and the distance from the hot material will determine the field of view (area being viewed by the scanner). The support should provide for rigid mounting free from vibration and free from flying objects, cobbles, etc. It must also remain at a temperature less than 135°F or be thermally insulated from the scanner mount.

Aim the scanner toward the area to be scanned by using the built-in front and rear sights located on top of the scanner housing. These sights are designed for aiming at a distance of 10 feet; therefore, allow slight adjustment for other distances. The point scanned will be within $\pm$ 4 inches of the center of the field of view at 10 feet when properly aimed using the sights. The long dimension of the field of view will be perpendicular to the axis of the cooling fittings located in the scanner body.

For fine adjustment of aiming, place an incandescent lamp or heat source at the desired aiming point. Connect the 115 VAC power to scanner terminals E and F. Connect an ohmmeter to terminals A and B. Turn power on, which will cause the chopper motor to rotate. Adjust horizontal and vertical position for minimum resistance and tighten mounting screws.

Control

The control panel (see outline diagram - Figure 4) may be mounted in a remote protected location. The cable run between control and scanner should be kept as short as possible but can be extended up to 1,000 feet, if necessary. Ambient temperature should not exceed 135°F at the control panel location.

INSTALLATION (continued)

ELECTRICAL CONNECTIONS

After the scanner and amplifier control have been permanently located and the scanner has been properly mounted and aimed, make wiring connections according to Figure 6. Plug the output relay or device into the octal socket. Figure 6 shows the power input connections and alternative functions of time delay and logic output. For the seal-in circuit, see Figure 7. If the 3S7505KS503 Static Switch is used, disregard output contact arrangement shown for the relay and refer to instructions supplied with the static switch.

Two 7/8-inch diameter access holes for the 1/2-inch water-tight flexible conduit fittings are provided in the scanner for separate wiring of the signal cable and the 115 VAC supply for the scanner chopper motor.

A separate run of the signal cable is recommended when using the static switch or logic output or where unusual noise (transients) may exist on the input AC supply lines. If a standard relay output is used and the AC line is relatively free from transients, both signal leads (properly shielded) and the 115 VAC motor leads may be run in the same conduit, separate from other power wiring.

Note requirement for the special high temperature scanner connection cable as listed in the section entitled DESCRIPTION when the ambient temperature exceeds 175°F.

If the mercury wetted-contact relay is used, refer to the instructions supplied with the relay for requirement of contact protection and vertical mounting.

COOLING REQUIREMENTS FOR THE SCANNER

CAUTION

COOLING MUST BE PROVIDED IF THE TEMPERATURE OF THE CELL-MOTOR PLATE CAN EXCEED 140°F. THIS WOULD NORMALLY BE EXPECTED FOR SURROUNDING AIR TEMPERATURES ABOVE 135°F; HOWEVER, IN ADDITION TO AIR TEMPERATURE, EXPOSURE TO RADIATION FROM LARGE AREA HIGH TEMPERATURE MATERIAL CAN ALSO INCREASE MOTOR AND CELL TEMPERATURES.

The scanner is equipped with 1/4-inch quick disconnect fittings suitable for either air or water cooling. For temperatures up to an equivalent of 180°F ambient, air cooling should be used. For temperatures up to an equivalent of 400°F, water must be used. Water cooling should not be used if air will provide sufficient cooling. For air cooling to temperatures equivalent to 180°F ambient, it is recommended that filtered shop air at 25 to 100 psi pressure and an inlet air temperature not exceeding 100°F be supplied to the scanner cooling chamber. The use of a Perfecting Service Company Catalog Number F100A or equivalent

INSTALLATION (continued)

COOLING REQUIREMENTS FOR THE SCANNER (continued)

quality air filter is recommended. The load on the air supply will be approximately 4.5 cu. ft./min. at 50 psi.

For water cooling, the minimum flow rate for temperatures equivalent to a 400°F ambient is 1/ 2 gallon per minute for a water temperature not exceeding 68°F and 1.0 gallon per minute for a water temperature of 86°F.

NOTE: Cooling requirements are based on ambient air temperature alone and do not take into account the heating which will be caused by radiant heat energy from the material being scanned. Also, note that the radiant energy may have little effect on air temperature, but at the same time can cause considerable heating of solid objects like the scanner. The equivalent temperature referred to in this section is the sum of the ambient air and the temperature increase due to radiation. The effects of the radiant energy can be greatly reduced by locating the scanner far from large hot materials and by installing polished heat shields or reflectors. If there is some doubt about the adequacy of the cooling means being used or if short scanner life is experienced, measure the temperature of the motor-cell mounting plate. This must not exceed 140°F and can be checked with a thermocouple wire and bridge. The couple should be placed under one of the screws which hold the motor-cell plate to the front casting. This check will only be valid if sufficient time is allowed and the normal flow of hot material is present during the measurement. If the 140°F temperature is being exceeded, check or increase cooling, relocate the scanner and/or install heat reflectors. Also, it may be necessary to replace the scanner since it may have been damaged by excess temperature.

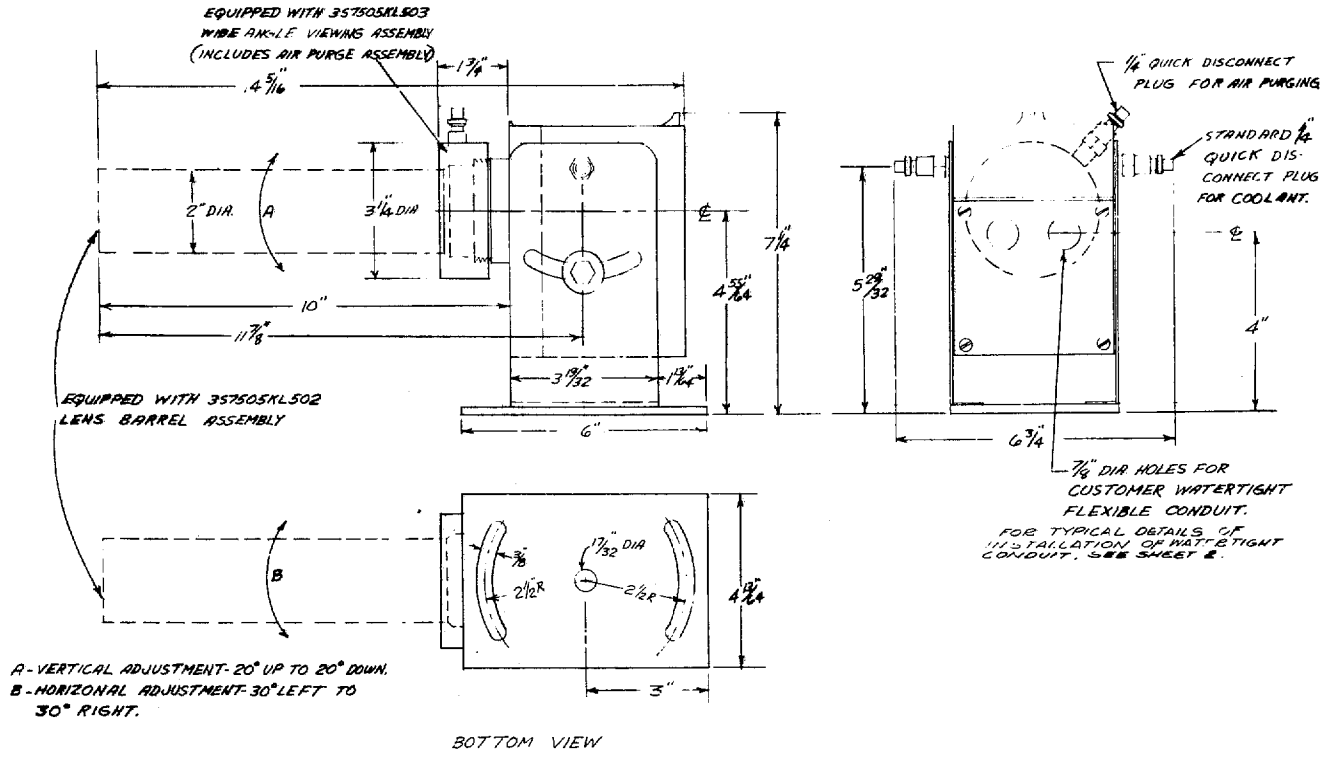
APPLICATION

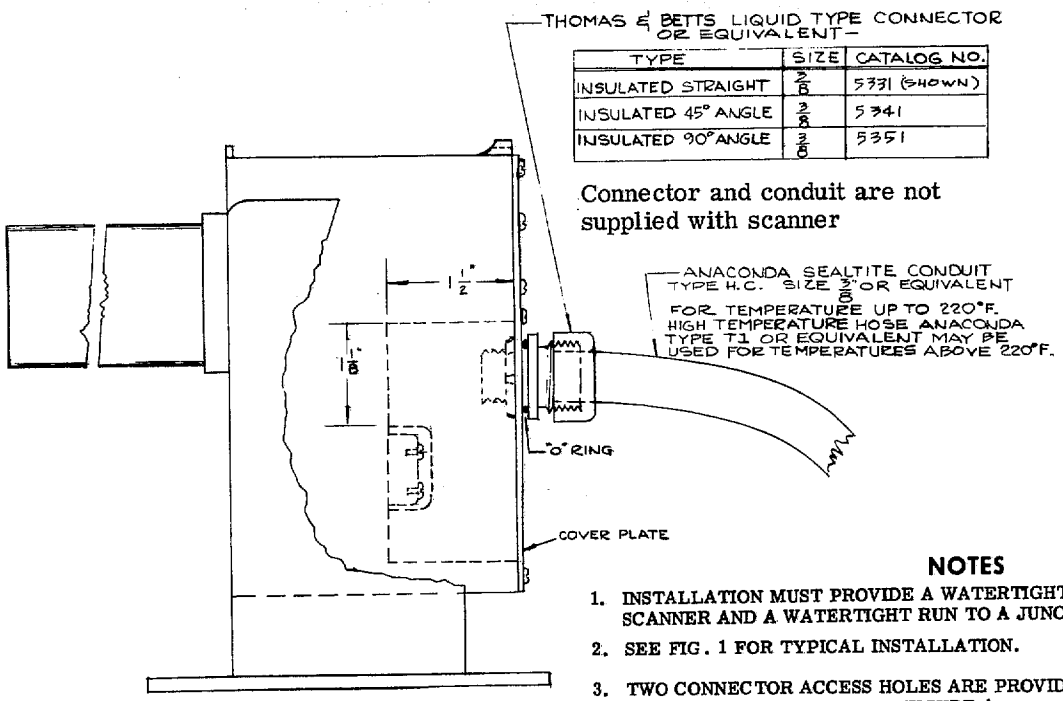
The following general information and three examples of typical applications may be used as a guide in selection of the best scanning assembly for a particular application. This information will also aid in making adjustments in selection of scanning arrangement if later changes in material, object temperature, etc., are made after the initial installation and operation.

The performance curves "A", "B", and "C" on pages 8 and 9 indicate the maximum background temperature allowable, and minimum object operating temperature for a wide range of emissivity times the percent field of view filled by the hot material. The three different curves are for the scanner body equipped with different scanning assemblies and apertures or window slots as listed under each curve.

All three curves are similar and for an explanation of usage, refer to curve "B". The maximum allowable background temperature is 800°F. Example: If the emissivity times the percent field of view filled is 20% and the minimum object temperature is 1750°F, then the maximum allowable background temperature is 800°F. Of course, in this example satisfactory operation is obtained with any background temperature below 800°F and for any object temperature above 1750°F.

APPLICATION (continued)





TYPE	SIZE	CATALOG NO.
INSULATED STRAIGHT	3/8	5331 (SHOWN)
INSULATED 45° ANGLE	3/8	5341
INSULATED 90° ANGLE	3/8	5351

3S7505HS500A6
RADIANT HEAT SCANNER

NOTES

1. INSTALLATION MUST PROVIDE A WATERTIGHT CONNECTION TO SCANNER AND A WATERTIGHT RUN TO A JUNCTION BOX.
2. SEE FIG. 1 FOR TYPICAL INSTALLATION.
3. TWO CONNECTOR ACCESS HOLES ARE PROVIDED IN TERMINAL BOARD COVER PLATE AS SHOWN ON FIGURE 4.

SEE INSTRUCTION BOOK GEH-2182, PAGE 8 "ELECTRICAL CONNECTIONS". IF ONLY ONE ACCESS HOLE IS USED, SEAL REMAINING HOLE WITH PLUG. G.E. CATALOG NUMBER CR104G41 OR EQUIVALENT.

Figure 5. Outline Diagram of 3S7505HS500 Scanner (44B208790 Rev. 3)

APPLICATION (continued)

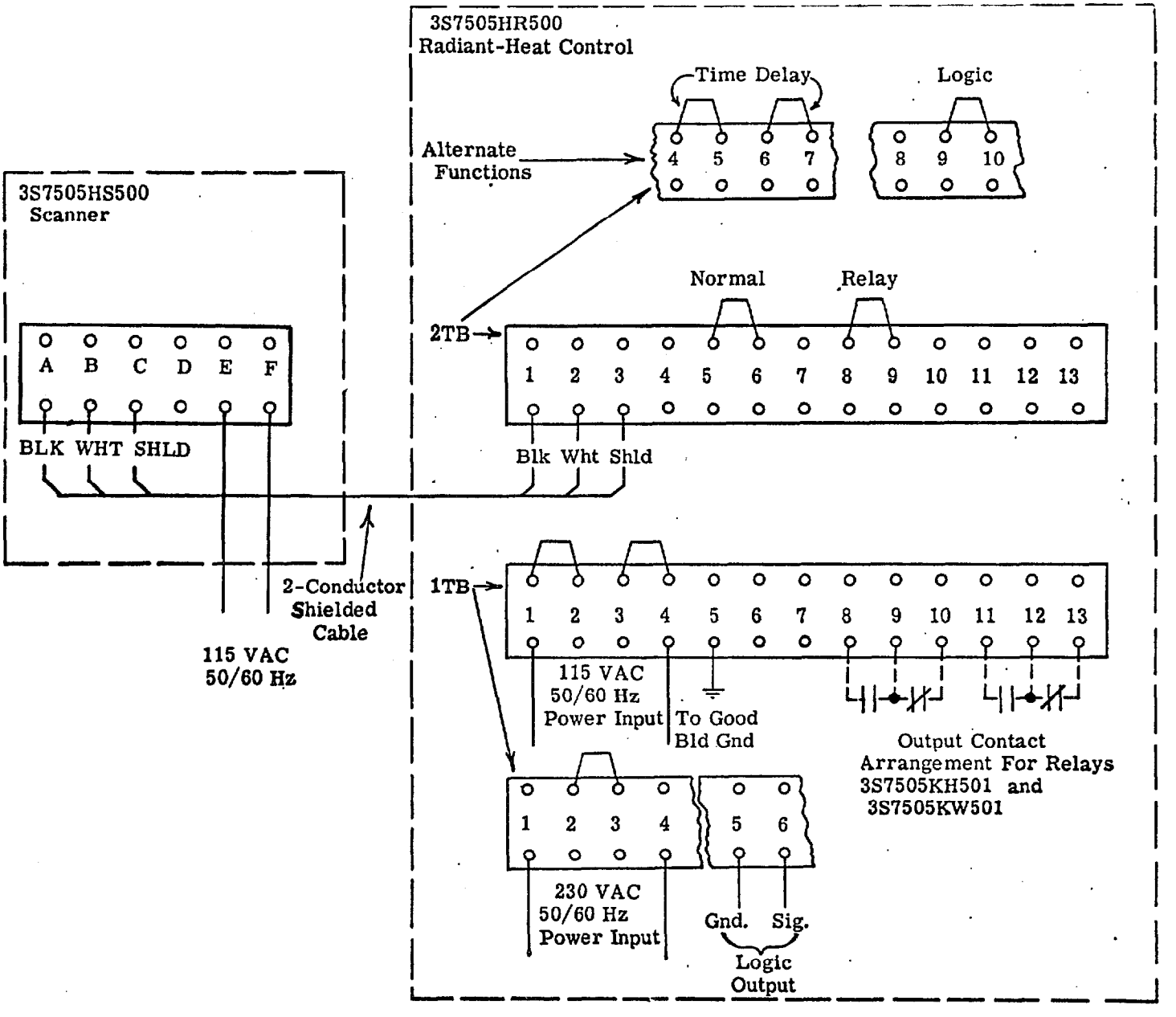


Figure 6. Customer Connection Instructions

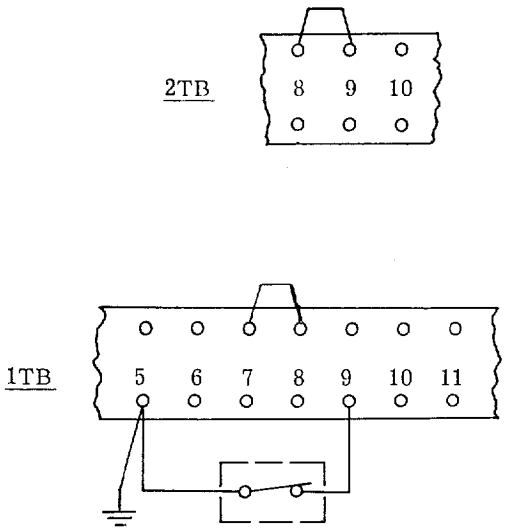


Figure 7. Connection For Seal-In (Open External Switch to Reset. One Set of Output Relay Contacts is Required.)

TABLE 2

RELATION OF COLOR TO TEMPERATURE OF IRON AND STEEL[1]

COLOR	DEG. F
Dark blood red, black red	990
Dark red, low red	1050
Dark cherry red	1175
Medium cherry red	1250
Cherry, full red	1375
Light cherry, light red	1550
Orange, free scaling heat	1650
Light orange	1725
Yellow	1825
Light yellow	1975
White	2200

[1] "Mechanical Engineer's Handbook", Lionel S. Marks, McGraw-Hill Book Company

APPLICATION (continued)

In curve "B", the minimum object temperature is 600°F and this is only permissible where the field of view is completely filled with the hot object of a material whose emissivity is one. Actually this is a theoretical value since most materials being detected never exceed 0.97 (see Table 3). At this minimum object temperature, the maximum background allowable would be 500°F.

(NOTE: The minimum object temperatures for various percentage lines is where the object temperature vertical line intersects the horizontal 500°F background temperature line.)

For a particular percentage line, the complete range of operating level of the object temperature is controlled by the setting of the sensitivity control. Example: note the 10% line; with the sensitivity control set at maximum, the minimum object temperature would be 950°F providing the background did not exceed 500°F. At the minimum sensitivity control setting, the minimum object temperature would be 1950°F with maximum allowed background of 800°F.

Generally, consider curve "A" with the 3S7505KL502 Lens Assembly and no aperture for low object and low ambient temperature conditions.

For medium range of object and background temperatures, and where the object size is relatively large, consider using curve "B" and the 3S7505KL502 Lens Assembly with the 0.175 in. diameter aperture. If the object size is small compared to the conveying bed, consider the 3S7505KL503 Wide Angle Scanning Assembly with one or the other window slots.

For high object and background temperatures which require the scanner to be mounted some distance away, consider curve "C" which uses the 3S7505KL502 Lens Assembly and the small diameter (0.031 in.) aperture.

Location of the scanner should provide for the best ratio of object radiated heat signal to the background radiation. It should also provide for the greatest percent of field of view to be filled with the hot object while still viewing the complete width of the conveying bed.

Items of Primary Considerations for Applying this Equipment

(1) Emissivity of material to be detected - see Table 3.

(2) Minimum object temperature to be detected.

(3) Size of object, width or height depending upon the aiming direction of scanner.

(4) Width of trough or conveying means.

(5) Expected maximum background temperature by radiation.

TABLE 3

EMISSIVITY OF MATERIAL (SURFACES)②

MATERIAL (Surface)	Temp* °F	Emissivity*
Aluminum		
Polished	440 - 1070	0.039 - 0.057
Oxidized at 1110°F	390 - 1110	0.11 - 0.19
Brass		
Oxidized at 1110°F	390 - 1110	0.61 - 0.59
Polished	494 - 710	0.033 - 0.037
Chromium	100 - 1000	0.08 - 0.26
Copper		
Heated at 1110°F	390 - 1110	0.57
Cuprous Oxide	1470 - 2010	0.66 - 0.54
Molten Copper	1970 - 2330	0.16 - 0.13
Gold		
Polished	440 - 1160	0.018 - 0.035
Iron		
Polished Pure Fe	300 - 1800	0.05 - 0.37
Ground Sheet Steel	1400 - 2000	0.52 - 0.60
Turned Cast Iron	70 - 1800	0.43 - 0.70
Oxidized Smooth	70 - 2000	0.80 - 0.90
Molten Cast Iron	2370 - 2550	0.29
Molten Mild Steel	2910 - 3270	0.28
Molybdenum		
Filament	1340 - 4700	0.096 - 0.292

*When two temperatures and two emissivities are given, linear interpolation is permissible. Temperatures below the minimum operating level of the equipment are listed for information and interpolation only and do not imply that this equipment will operate from any temperatures below those specified in performance curves "A", "B", and "C".

②"Mechanical Engineer's Handbook", Lionel S. Marks, McGraw-Hill Book Company.

TABLE 3 (CONTINUED)

EMISSIVITY OF MATERIAL (SURFACES)②

MATERIAL (Surface)	Temp* °F	Emissivity*
Monel Metal		
Oxidized at 1110°F	390 - 1110	0.41 - 0.46
Nickel		
Pure Polished	70 - 700	0.045 - 0.087
Wire	368 - 1844	0.096 - 0.186
Plate (Oxidized at 1110°F)	390 - 1110	0.37 - 0.48
Oxide	1200 - 2290	0.59 - 0.86
Chromnickel	125 - 1894	0.64 - 0.76
KA-2S Alloy (8Ni, 18Cr)		
Oxidized	420 - 914	0.44 - 0.36
NCT-3 Alloy (20Ni. 25Cr)		
Oxidized	420 - 980	0.90 - 0.97
NCT-6 Alloy (60Ni. 12Cr)		
Oxidized	520 - 1045	0.89 - 0.82
Platinum		
Polished Plate	440 - 2960	0.054 - 0.17
Filament	80 - 2240	0.036 - 0.192
Silver		
Pure Polished	440 - 1160	0.20 - 0.032
Tantalum - Filament	2420 - 5430	0.194 - 0.33
Titanium	1490 - 2460	0.459 - 0.484
Tungsten		
Aged Filament	80 - 6000	0.032 - 0.018

*When two temperatures and two emissivities are given, linear interpolation is permissible. Temperatures below the minimum operating level of the equipment are listed for information and interpolation only and do not imply that this equipment will operate from any temperatures below those specified in performance curves "A", "B", and "C".

②"Mechanical Engineer's Handbook", Lionel S. Marks, McGraw-Hill Book Company.

TABLE 3 (CONTINUED)

EMISSIVITY OF MATERIAL (SURFACES)②

MATERIAL (Surface)	Temp* °F	Emissivity*
Zinc Oxidized at 750°F	750	0.11
Asbestos	100 - 700	0.93 - 0.95
Candle Soot Lampblack Water Glass	70 - 700	0.95

Items of Primary Considerations for Applying this Equipment (continued)

(6) Available location of scanner; this will determine the field of view for the scanning assembly to be used.

(7) Select smallest aperture as determined by curves "A", "B", and "C" which will give satisfactory operation. This will prevent overheating of detector due to radiant energy from the hot object remaining in the field of view.

Typical Application Number 1

In a rod mill, the requirement is to detect the presence of steel rods with varying diameters of 1/4 in. to 1/2 in. traveling in a 6 in. wide trough. The rods have a smooth oxidized surface.

Conditions Given

(1) Emissivity of steel - use 0.90 per Table 3, page 19.

(2) Rod temperature - minimum 1600°F.

(3) Location of scanner - 3 feet above trough.

*When two temperatures and two emissivities are given, linear interpolation is permissible. Temperatures below the minimum operating level of the equipment are listed for information and interpolation only and do not imply that this equipment will operate from any temperatures below those specified in performance curves "A", "B", and "C".

②"Mechanical Engineer's Handbook", Lionel S. Marks, McGraw-Hill Book Company.

APPLICATION (continued)

Determine the scanning assembly required, the percent field of view filled and the maximum background temperature allowed for satisfactory operation.

(a) Since the object size is small and the temperature is mid-range, refer to curve "B" and try the wide-angle scanning assembly 3S7505KL503 with the larger window slot (0.06 in. x 0.167 in.).

(b) The percent field of view filled with the hot object is determined as follows: From the specification, the field of view of the window selected above is 0.7 in. x 2.0 in. per ft. Since the scanner is located 3 feet from the trough and mounted with the long axis of the window perpendicular to the direction of rod travel, the viewed width is 3 feet x 2 in./ft. = 6 in.

Since it is necessary to detect the smallest rod diameter which is 1/4 in., then the percent field of view filled is $\frac{0.25 \text{ in.}}{6 \text{ in.}} = 4.1\%$.

(c) The emissivity of the steel rod is given as 0.90; therefore, the percent field of view filled (4.1%) times the emissivity (0.90) equals 3.7%.

(d) Refer to curve "B" and note that the 3.7% line will fall between the 2% and 5% lines. Note also the intersection of the 1600°F object temperature line (vertical) falls within the satisfactory operation range with a maximum allowable background temperature of 600°F. The sensitivity control will be set at approximately 2/3 maximum range (between 6 and 7 on the dial).

Since the 1/2-inch diameter rod would result in a greater percent field of view filled, it would provide even greater ratio of signal to background and satisfactory operation is assured.

Some observations can be made on the above application if any adjustments are required in scanner mounting, etc.

(1) If the background temperature would never exceed 500°F and the sensitivity control set toward maximum (10), satisfactory operation could be expected even if the rod temperature dropped to 1200°F.

(2) The scanner could not be moved any closer using the selected window above and still scan the complete width of the trough, because its field of view would be less than the 6 inches.

(3) If the scanner were moved farther away, its percent field of view filled would be reduced, and the background temperature allowed would be less and the minimum object temperature required would be higher.

(4) If the 0.01 in. x 1 in. window slot had been selected, the percent field of view filled would be very small since the field viewed by the scanner would be 10 in./ft. by 3 ft. or 30 in.; thus, only 0.25 in./30 in. or 0.8%.

APPLICATION (continued)

(5) If the standard lens assembly with the 0.175 in. aperture were used, it would have only seen 3 in. or 1/2 of the trough width since its field of view is 1.0 in./ft. However, if the scanner could have been located 6 feet away, then this would have been a satisfactory selection.

Typical Application No. 2

A requirement to detect the presence of copper strips in a finishing mill.

Conditions Given

(1) Temperature of copper at point of scanning - 1000°F.

(2) Width of strip - 10 in.

(3) Width of conveyor trough - 36 in.

(4) Emissivity factor - use 0.57 from Table 3.

(5) Location of scanner - desired 3 feet above and 1 foot from the side of the trough.

Determine the scanning assembly and/or window slot and background temperature allowed for satisfactory operation.

(a) Object temperature is mid-range and width of trough will require the wide angle scanning assembly since the scanner is mounted fairly close. Consider using the 3S7505KL503 assembly with the 0.01 in. x 1 in. window slot and reference curve "B".

(b) The percent field of view of the scanner will be 10 in./ft. x 3 ft. or 30 in. (NOTE: This will not scan the complete width of the trough; therefore, the scanner should be moved farther away to 3 1/2 ft. This will allow the scanner to view 35 in. width.) Since the strip width is 10 in. and the scanner is offset 1 foot from the side of the trough, the additional inch of scanning required will be ignored. The percent field of view filled will now be 10 in./35 in. = 29%.

(c) The emissivity (0.57) times the percent field of view filled (29%) equals 16.5%.

(d) Refer to curve "B" and note that the 16.5% line will fall approximately midway between the 10% and 20% lines.

(e) Note the intersection of the 1000°F object temperature vertical line with the 16.5% line. This falls within a satisfactory operating range with a maximum allowable background temperature of 520°F. The sensitivity control will be set near the maximum (9 to 10 on dial).

APPLICATION (continued)

Typical Application No. 3

Requirement to detect the absence of any steel billets on a run-out table.

Conditions Given

(1) Minimum temperature of billet at scanning point - 800°F.

(2) Table width - 10 ft.

(3) Billets may vary 3 in. to 10 in. high.

(4) Scanner to be located 3 ft. from the side of the table and looking downward 5° from horizontal.

(5) Emissivity factor - use 0.90 from Table 3.

Determine the scanning assembly required, maximum allowable background temperature, and if satisfactory operation can be obtained from the 800°F billet temperature.

(a) Try the 3S7505KL502 Lens Assembly with no aperture and refer to curve "A" on page 8.

(b) The percent field of view filled is the field of view of the lens system times the distance which is 1 in./ft. x (10 ft. + 3 ft.) = 13 in. The scanner is mounted with the axis of cooling fittings horizontal. Since the minimum height billet is 3 in., the percent field of view filled is 3 in./ 13 in. = 23%.

(c) Emissivity (0.90) times the percent field of view filled (23%) equals 20.7%. Curve "A", looking at the point where the 800° object temperature line intersects the 20% line, indicates that satisfactory operation can be obtained at the sensitivity control setting of about 1/3 range (between 3 to 4 on the dial). Maximum background temperature allowed is 400°F.

OPERATION

After the scanner and control have been properly mounted, aimed, and wiring connections made, apply power to both units. With the scanner viewing the background area (no hot object in view), adjust the sensitivity control toward maximum and note the setting where the relay just deenergizes (indicator light off). If the relay does not drop out at maximum setting, this is satisfactory and indicates that the level of radiation from the background is not sufficient to operate the control. Next, move the hot object into the scanner field of view (the relay should drop-out and the indicator light should turn off) and adjust the sensitivity control toward minimum to the point where the relay just energizes;

OPERATION (continued)

note this setting. If the relay does not energize at the minimum setting, this is satisfactory and indicates that enough radiation is available from the hot material to operate the control at minimum sensitivity. Finally, set the sensitivity control to the mid-point of the two settings noted above. This procedure will provide optimum setting of the sensitivity control which will insure adequate signal margin above the minimum required and still not allow operation from the background radiation.

The above procedure assumes that the background temperature was up to its normal operating temperature; readjustment of the sensitivity control may be required if the background ambient increases excessively.

If the time delay function is used, adjust the time delay control for the desired time before relay energizes upon removal of heat signal.

PRINCIPLES OF OPERATION

The 3S7505HS500 Scanner consists of a heat sensing lead-sulfide detector cell and a synchronous motor-driven chopper disc containing 90 slots. The motor rotates at approximately 1,800 RPM which results in a modulated heat signal at a frequency of approximately 2.7 KHz.

The modulated heat signal is connected to the 3S7505HR500 Control which contains the power supply, associated transistorized circuitry to amplify the signal, and the output relay or alternate device.

Refer to Figure 8 for the typical elementary diagram. The signal level is controlled by the setting of sensitivity potentiometer 1P and coupled through 1C to three stages of amplification consisting of 1Q through 6Q. It is then rectified and applied as a positive signal to the base of 7Q (unless time delay is used). When a sufficient heat signal is present on 1PD, in the scanner, 7Q is turned on causing 14Q to conduct which turns off 15Q, thus indicator light 1L is turned off. Also, when 7Q is turned on, 8Q conducts and 9Q is turned off, deenergizing the output relay 1CR or the alternate output device.

If the amplifier control is connected for logic output and the heat signal turns 7Q on, 9Q is turned off resulting in a positive logic voltage level of 24 volts at 1TB(6). Conversely, when the hot object is not present, 9Q conducts and the logic output at 1TB(6) goes to zero.

If the time delay function is used, the positive signal (hot object in scanner view) at 2TB(5) is jumpered to 2TB(4) which causes 13Q, 10Q, and 11Q to conduct and charge timing capacitor 12C. When the heat signal is removed, the negative charge on 12C keeps 12Q off and 7Q on, thus relay 1CR remains deenergized for an adjustable time. The adjustable time delay is controlled by the time delay potentiometer 2P. After 12C discharges, 12Q conducts, turning off 7Q and relay 1CR energizes.

MAINTENANCE

This equipment requires very little maintenance beyond the removal of any accumulation of dirt or dust on the scanner lens. The extension barrel may be unscrewed from the lens assembly for accessibility to the lens for cleaning.

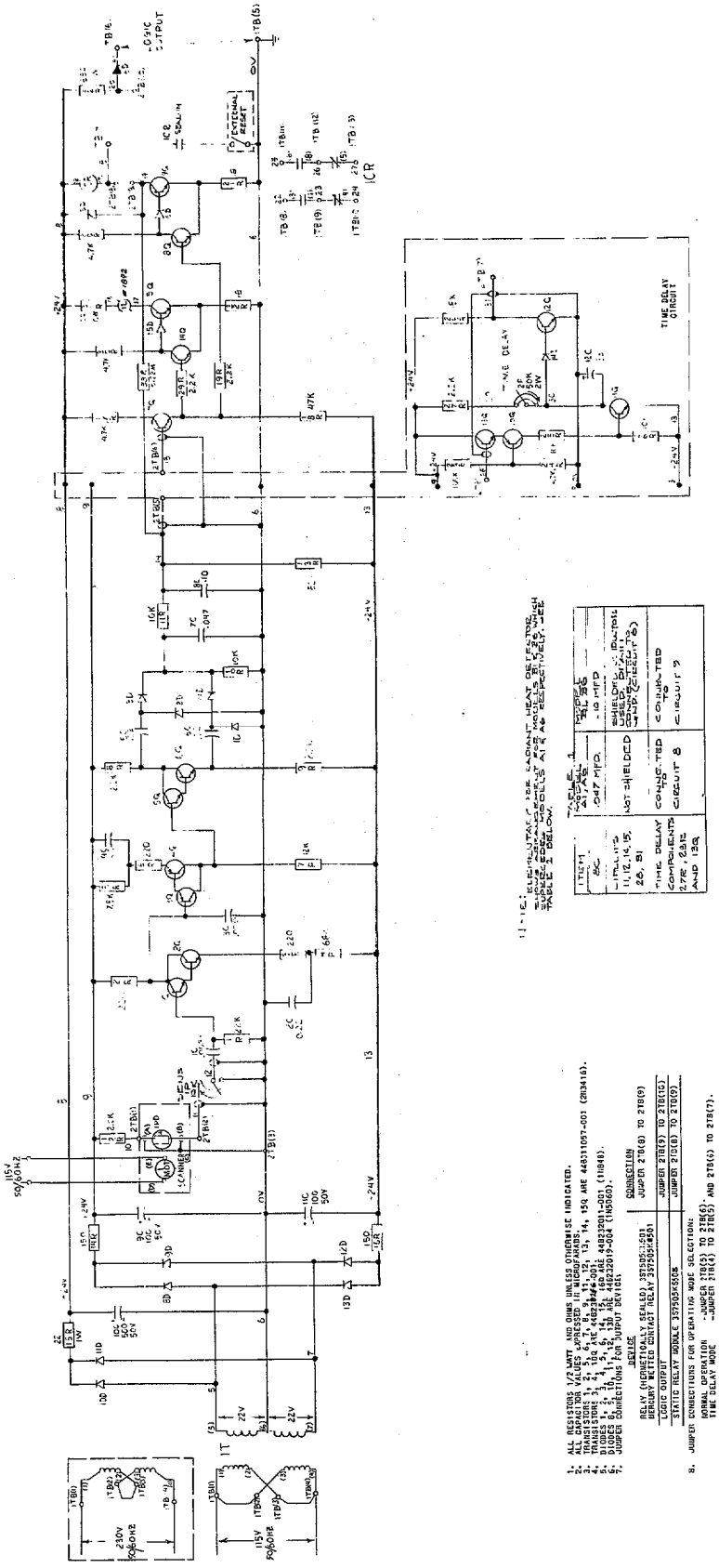
For access to the motor and cell assembly in the scanner, disconnect power and remove the four screws holding the terminal board cover plate in place. Then disconnect the 3 motor leads (black, blue and red) and the 3 cell leads (shielded cable). Finally, loosen the four retaining screws holding two casting sections together and carefully separate the two sections while feeding the motor and cell wires through the holes in the main casting. Note: Silicone lubricant has been applied to the mating surfaces of the motor mounting plate and cooling chamber to insure good heat transfer; do not remove this lubricant.

For access to the slotted disc, remove the two flat head screws securing the mounting plate to the front casting section.

The motor, cell or disc may be replaced if necessary; however, it is recommended that a complete motor and cell assembly be installed. See Renewal Parts List for catalog numbers. If the disc assembly is moved or replaced on the motor shaft, 1/32-inch clearance must be provided between the disc and cell face.

RENEWAL PARTS

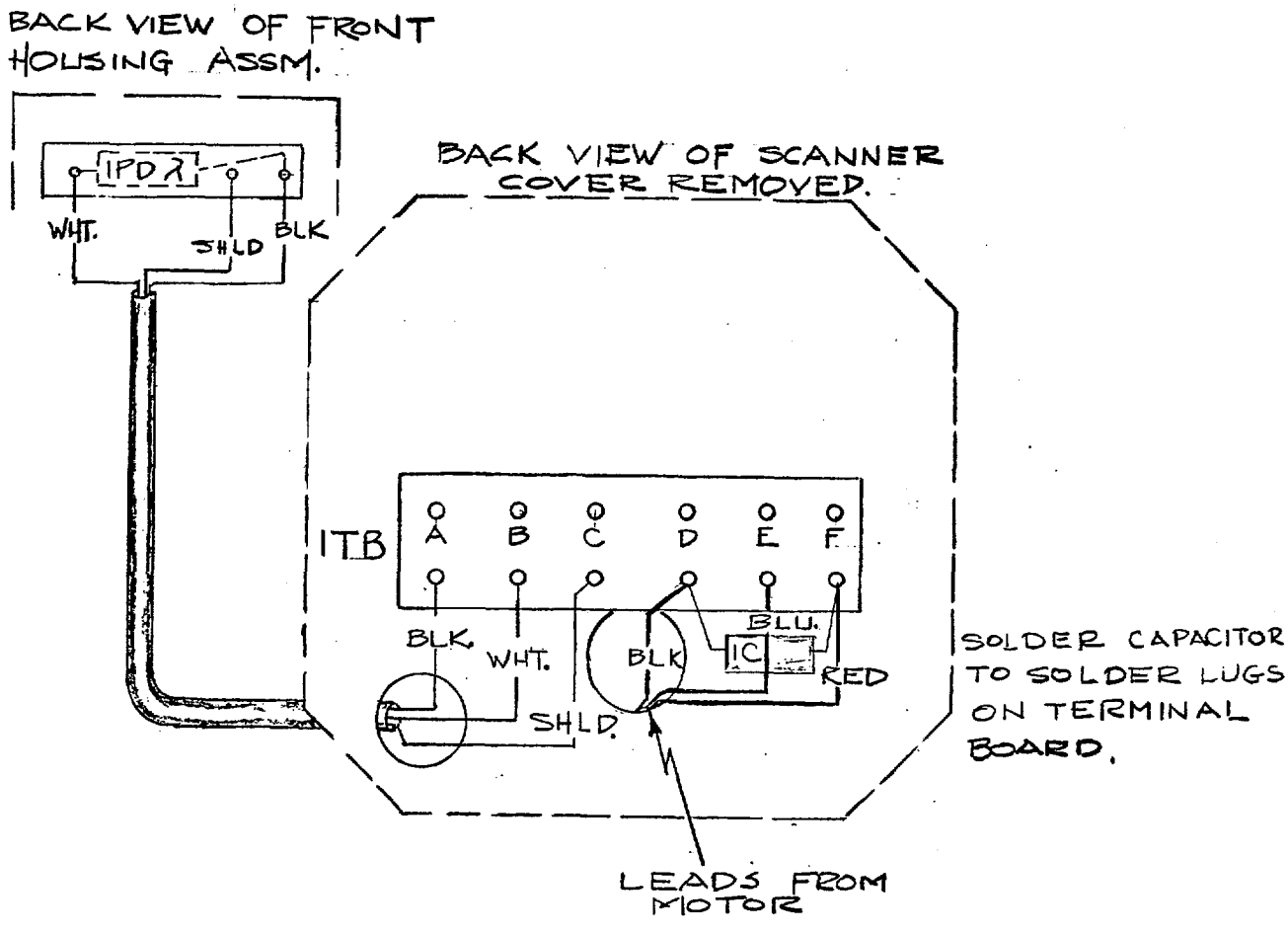
When ordering renewal parts, address the nearest sales office of General Electric Company, specify the quantity required, and give the catalog numbers or describe the required parts in detail. In addition, give the 3S number and the complete nameplate rating of the unit.



MODE OF OPERATION

Heat Signal	Relay	Light	Logic
Present	De-energized	Off	+24
Absent	Energized	On	0

Figure 8. Elementary Diagram of 3S7505HR500 Radiant-Heat Detector System (44D300097 Rev. 2)



NOTE.

LEADS SUPPLIED WITH MOTOR.
SHIELDED 2-COND CABLE 44A316848-003 USED FOR CELL CONNECTIONS.

Figure 9. Connection Diagram of 3S7505HS500 Scanner (44A308808)

PRINCIPAL RENEWAL PARTS LIST
FOR: 3S7505HR500 RADIANT-HEAT DETECTOR CONTROL PANEL

DIAGRAM SYMBOL	QTY. IN USE B1, B6		PART NUMBER	DESCRIPTION
	1		3S7505KF138B1	Printed Circuit Board
1, 3C		2	K8277646F33	Capacitor .0033 mfd, 600V
2, 4, 5, 6C		4	933B597H22	Capacitor .22 mfd, 100V
7C		1	933B597G47	Capacitor .047 mfd, 100V
8C		1	933B597H10	Capacitor .1 mfd, 100V
9, 11C		2	K9774719L10	Capacitor 100 mfd, 50V
10C		1	K9774719L50	Capacitor 500 mfd, 50V
12C		1	44B310008-K33	Capacitor 33 mfd, 25V
1, 2R		2	933B590G22	Resistor 22K ohm, 1/2 W
3, 6R		2	933B590E22	Resistor 220 ohm, 1/2 W
4R		1	933B590G68	Resistor 68K ohm, 1/2 W
5R		1	933B590F75	Resistor 7.5K ohm, 1/2 W
7R		1	933B590G12	Resistor 12K ohm, 1/2 W
8,9,12,19,27,29R		6	933B590F22	Resistor 2.2K ohm, 1/2 W
10, 11, 25, 26R		4	933B590G10	Resistor 10K ohm, 1/2 W
13R		1	933B590G82	Resistor 82K ohm, 1/2 W
14, 16R		2	933B590E15	Resistor 150 ohm, 1/2 W
15R		1	M6986159D22	Resistor 22 ohm, 1 W
17, 20, 24, 30R		4	933B590F47	Resistor 4.7K ohm, 1/2 W
18R		1	933B590G47	Resistor 47K ohm, 1/2 W
21, 32R		2	933B590D18	Resistor 18 ohm, 1/2 W
22R		1	M6986159E68	Resistor 680 ohm, 1 W
23R		1	933B590H10	Resistor 100 K ohm, 1/2 W
28R		1	933B590G18	Resistor 18K ohm, 1/2 W
31R		1	K8622212E10	Resistor 100 ohm, 10 W
33R		1	933B590H22	Resistor 0.22 megohm, 1/2 W
1,2,3,4,5,6,14 15, 16D		9	44B232011-001	Diode
8 through 13D		6	44B232019-004	Rectifier, 1A
1,2,5,6,7,8,9,11 12,14,13,15Q		12	44B311057-001	Transistor
3, 4, 10Q		3	44B238246-001	PNP Silicon Transistor
1T	1		44B213365-G01	Transformer
	1		M9664579P1	Socket
	1		44B235850-001	Indicator Light
	1		1892	Lamp
1P	1		M9729433G10	Potentiometer 10000 ohm, 2 W
2P	1		44A337916-110	Potentiometer 50000 ohm, 2 W
	1		44A317700-001	Indicator Lens

PRINCIPAL RENEWAL PARTS LIST
FOR: 3S7505HS500 RADIANT-HEAT SCANNER

DIAGRAM SYMBOL	QTY. IN USE C6	PART NUMBER	DESCRIPTION
	1	44A373494-G01	Cell and Motor Assembly
	1	44A373379-001	Motor (50/60 Hz)
	1	44A317670-G02	Cell Assembly
	1	44A373876-G01	Disc Assembly
	1	44B211519-059	Terminal Board
	1	44A370868-J10	Capacitor 1.0 mfd, 400V

PRINCIPAL RENEWAL PARTS LIST
FOR: 3S7505KL502A1 LENS BARREL ASSEMBLY

DIAGRAM SYMBOL	QUANTITY	PART NUMBER	DESCRIPTION
	1	3S7505KL502A1	Lens Barrel Assembly
	1	44A317663-G01	Lens Barrel Assembly
	2	K9381478P1	Lens
	2	M9605846P33C	Retaining Ring
	2	V9092476P1	Gasket
	1	44A212187-009	"O" Ring
	1	44A316093-004	Aperture
	1	44A316093-005	Aperture

PRINCIPAL RENEWAL PARTS LIST
FOR: 3S7505KL503A1 WIDE ANGLE SCANNING ASSEMBLY

DIAGRAM SYMBOL	QUANTITY	PART NUMBER	DESCRIPTION
	1	3S7505KL503A1	Wide Angle Scanning Assembly
	1	44A317906-G01	Barrel-Window Assembly
	1	V9092476P1	Gasket
	1	K9385297P1	Window
	1	M9605846P33C	Retaining Ring
	1	44A212187-009	"O" Ring
	1	3S7505KF139A1	Air Purge Collar
	1	44A317902-G01	Aperture
	1	44A317902-002	Aperture

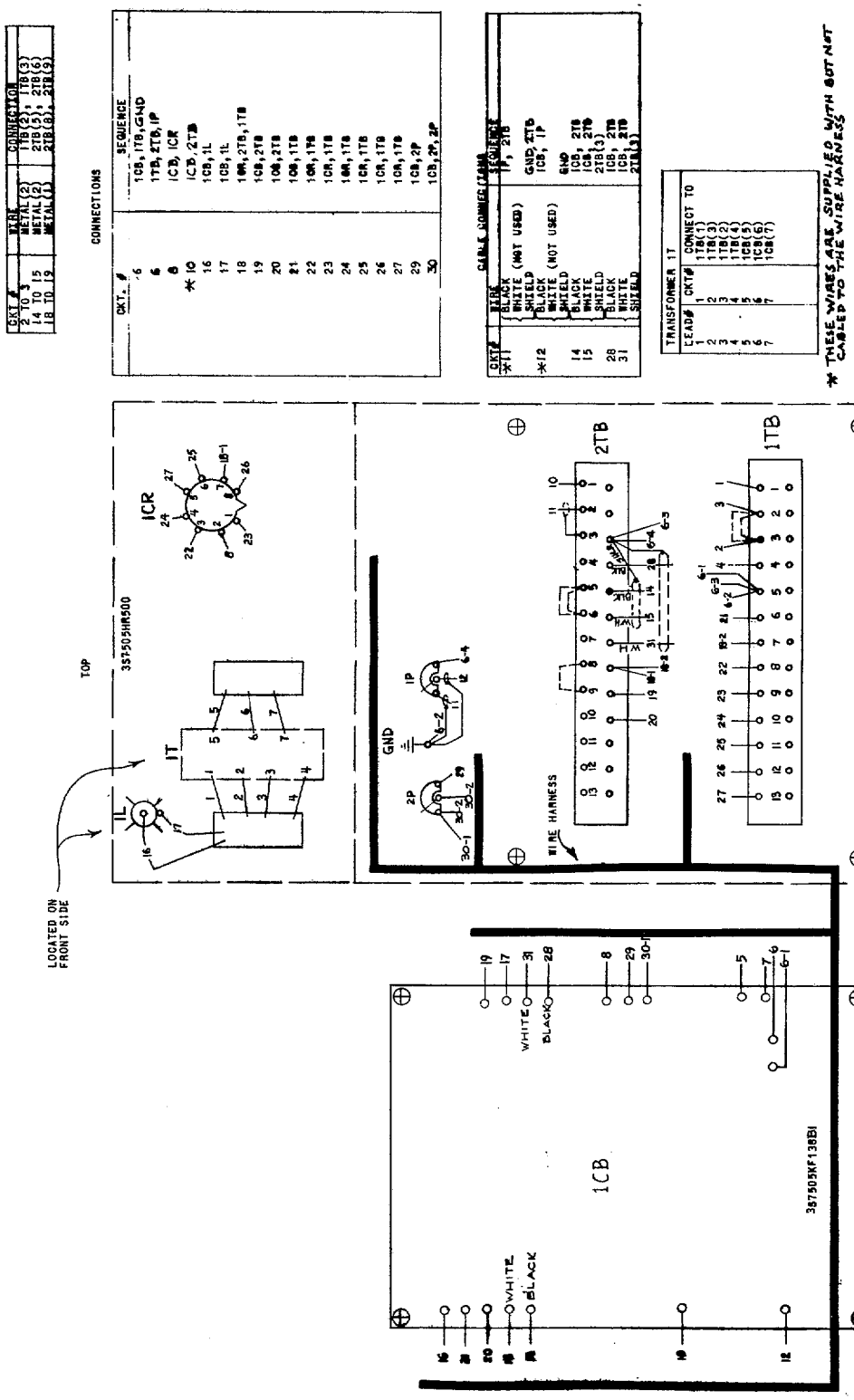


Figure 10. Connection Diagram of 387505HR500 B1, B6 Control (44C306538)

Note: For Models A1, A6 obtain Diagram 44C300360 from the factory.

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